On Optimality of Local Maximum-Likelihood Detectors in Large-Scale MIMO Channels

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***Abstract*— The replica method originated from statistical mechanics has been successfully applied to analyzing performance of the global maximum-likelihood (GML) MIMO detector in the large-system limit. In this paper, the analysis is extended to the local maximum-likelihood (LML) detectors. A bit error rate (BER) formula for the LML detectors with a fixed neighborhood size is obtained by the replica method and interestingly by the method of Gaussian approximation as well. It is shown that the LML BER is always one of the solutions to the GML BER in any system configuration. Furthermore, the LML BER is the only solution of the GML BER in a broad range of system parameters of practical interest. In the high signal-to-noise ratio regime, both LML and GML detectors achieve the AWGN channel performance when the channel load is up to 1.51 bits/dimension with an equal-energy distribution, and the load can be higher with an unequal-energy distribution. This analytical result is verified by simulation that the sequential likelihood ascent search detector, which is a linear-complexity LML detector, can approach the BER of the NP-hard GML detector predicted by the analysis. This result might be practically useful in large MIMO systems.**

***Index Terms*— Massive MIMO, replica method, global maximum likelihood, local maximum likelihood.**

I. Introduction

In a multiple-input multiple-output (MIMO) channel a number of dada symbols each modulated by a random channel vector are transmitted. The receiver receives a superimposed version of the symbol-modulated vectors and demodulates the data symbols by using some specific detection scheme. The performance of various detectors has been extensively studied in the literature. The results for finite-size systems are in general very complex and provide little insight. On the other hand, the performance analysis can be greatly simplified in the large-system limit where both the number of symbols and the dimension of channel vectors tend to infinity with their ratio fixed. It is found that the fluctuation of detection performance due to the randomness of channel vectors and noise vanishes in the large-system limit [2]–[4].

The replica method originally developed in statistical physics has proven to be a powerful tool in the asymptotic analysis of large MIMO systems. Tanaka was the first to use the replica method to derive the bit error rate (BER) of the global conditional mean (GCM) detector (or marginal posterior mode detector), multiuser efficiency and spectral efficiency of a large random-spreading CDMA system with uncoded equal-energy BPSK signals [5]; in particular, a BER formula for the jointly optimum detector is obtained. Guo and Verdu presented a unified treatment of Gaussian CDMA channel and multiuser detection with an arbitrary energy distribution [6]–[8], and extended the results to multicarrier CDMA and frequency-selective fading channels [9]. Muller and Gerstacker studied the capacity loss due to the separation of detection and decoding [10]. More recently, Takeuchi *et al.* presented a large-system analysis of joint channel estimation, conditional mean detector, and per-user decoding for MIMO CDMA systems [11]. In addition, the replica method has been applied to the asymptotic analysis of MIMO systems with a large number of transmit and receive antennas. Moustakas derived the statistics of the mutual information of a single-user MIMO system in the presence of spatial correlation and interference [12]. Wen and Wong studied the achievable sum-rate of MIMO multiaccess system in a spatially correlated Rayleigh fading environment [13] and later extended the result to the Rician fading channel [14]. Takeuchi derived lower bounds on the achievable rates of successive decoding receivers in the large-system limit. The channel estimation overhead and channel estimation error are taken into account in the analysis [15]. Recently some results of replica analysis are corroborated by the works on large-dimensional random matrix theory [16].

The jointly optimum MIMO detector chooses the symbol vector that achieves the *global* maximum likelihood (GML) among all possible vectors. In contrast, a *local* maximum-likelihood (LML) detector [17], which the GML detector is akin to, selects a vector that attains the maximum likelihood among the vectors in its neighborhood. By specifying the size of neighborhood, an LML detector can achieve a certain tradeoff between the performance and the computational complexity. The LML detector with the smallest neighborhood size can be efficiently implemented by the wide-sense sequential likelihood ascent search (WSLAS) detector [18] and its average per-bit computational complexity is linear in the total

Manuscript received June 3, 2016; accepted July 24, 2016. Date of publication July 29, 2016; date of current version October 7, 2016. This work was supported in part by the National Natural Science Foundation of China under Grant 61571120 and Grant 61329101, in part by the National Basic Research Program of China (973 Program) under Grant 2013CB329203, in part by the Leading Talents Program of Guangdong Province under Grant 00201510, and in part by the Basic Research Program of Shenzhen under Grant JCYJ20151117161854942. The associate editor coordinating the review of this paper and approving it for publication was J. Zhang.

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Digital Object Identifier 10.1109/TWC.2016.2596721

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number of bits [18]. Simulation results showed that the linear-complexity WSLAS detector can approach the BER of the NP-hard GML detector in a region of practical large CDMA systems [19] and approach the BER of single-bit bound (SBB) in the high signal-noise ratio (SNR) regime. While the large-system GML BER formula is obtained by the replica analysis, little is known yet about the LML BER and its relationship to the GML BER in the literature.

In this paper, we intend to obtain a large-system BER formula for the LML detector by extending the replica analysis of the GML detector in [5]–[7]. The extension is not obvious at the first sight. To circumvent this, we first define the *local* conditional mean (LCM) detector with any neighborhood size that includes the LML detector as an instance. When the neighborhood size is the maximum, the LCM detector becomes the GCM detector and accordingly the LCM detector can be analyzed by the replica method in the similar way for analysis of the GCM detector. The main difference between the analyses of the LCM and GCM detectors is that the range of subshell for the LCM detector is general for all neighborhood sizes and include that for the GCM detector as a special instance. We then focus on the LCM detector with a fixed neighborhood size in the large-system limit. As a particular instance of the LCM detector, the LML detector is analyzed and an LML BER formula is obtained. It is surprising that the same LML BER formula is also obtained by the conventional method of Gaussian approximation. Several optimality properties of the LML detector are further revealed. It is found that the LML BER is always one of the solutions to the GML BER, including the low GML BER. In the high SNR regime, like the GML detector the LML detector can achieve the AWGN BER in a broad range of channel load of practical interest. Simulation results verify that for the same range of channel load, the sequential likelihood ascent search (SLAS) detector, which is a linear-complexity LML detector using only local information, achieves the NP-hard GML BER.

The remainder of the paper is organized as follows. In Section II, the channel model is presented and the LML, GML, LCM, and GCM detectors are defined. In Section III, the LML BER formula is derived first by the replica method and then by the method of Gaussian approximation. Section IV studies the optimality of the LML detector in comparison with the GML detector, the achievable channel load, and the achievability of analytical LML/GML BER by the SLAS detector in practical systems. Conclusions are given in Section V and some proofs are given in the Appendix.

II. Preliminaries

A. The Channel Model

Consider a general MIMO channel where K complex symbols are transmitted through a complex Gaussian channel. Each symbol is equiprobably selected from an alphabet. The kth symbol b_k is modulated by an N-dimensional complex channel vector $\mathbf{s}_k$. The energy of the kth symbol is A_k^2. The receiver receives a superimposed signal of symbol-modulated channel vectors plus noise

$$\mathbf{r} = \mathbf{SAb} + \mathbf{m} \tag{1}$$

where $\mathbf{S} = (\mathbf{s}_1, \cdots, \mathbf{s}_K), \mathbf{A} = \mathrm{diag}(A_1, \cdots, A_K), \mathbf{b} = (b_1, \cdots, b_K)^T$, and the noise $\mathbf{m} \sim \mathcal{CN}(\mathbf{0}, \sigma_0^2 \mathbf{I}_N)$.

The probability density function of $\mathbf{r}$ given $\mathbf{b}, \mathbf{S}, \mathbf{A}$ is

$$p_0(\mathbf{r}|\mathbf{b}, \mathbf{S}) = \frac{1}{(\pi\sigma_0^2)^N} \exp\left(-\frac{1}{\sigma_0^2}\|\mathbf{r} - \mathbf{SAb}\|^2\right) \tag{2}$$

where $\|\cdot\|$ is the l_2 norm. A sufficient statistic can be obtained from the matched filter (MF) $\mathbf{S}^H$ as $\mathbf{y} = \mathbf{S}^H\mathbf{r} = \mathbf{RAb} + \mathbf{n}$ where $\mathbf{R} = \mathbf{S}^H\mathbf{S}$ is the crosscorrelation matrix of channel vectors and $\mathbf{n} = \mathbf{S}^H\mathbf{m} \sim \mathcal{CN}(\mathbf{0}, \sigma_0^2\mathbf{R})$ with H denoting the Hermitian transpose. We will use both $\mathbf{r}$ and $\mathbf{y}$ in the rest of paper whenever one is more suitable.

The channel model (2) represents several types of communication systems. For example, a number of users each equipped with one or multiple transmit antennas access a basestation (BS) that is equipped with N receive antennas. As another example, in a CDMA system where K users access a BS with one antenna, $\mathbf{s}_k$ represents the spreading sequence of the kth user [1]. Moreover, by the multicode technique that multiplexes multiple symbols with extended symbol periods, each user can transmit multiple symbols no matter whether the spectrum is spread and whether there is one or multiple users [19]. The multi-transmit antenna, multi-receive antenna, CDMA, and multicode techniques can be applied in a combination and each yields the same channel model (2).

B. Local and Global Maximum-Likelihood Detectors

A number of detectors can be employed by the receiver to demodulate the symbol vector $\mathbf{b}$ from the received signal $\mathbf{r}$. We are particularly interested in the LML detectors and their performance in the large-system limit. An LML detector is dependent on the neighborhood size of a symbol vector. The neighborhood of a symbol vector $\mathbf{b}$ with neighborhood size J is defined by the set of vectors that differ from $\mathbf{b}$ by at most J symbols

$$\Omega_J(\mathbf{b}) = \{\mathbf{a} \in \Omega_K : \mathbf{a} \text{ differs from } \mathbf{b} \text{ by at most } J \text{ elements}\}$$

where Ω_K is the set of all possible symbol vectors. Given $\mathbf{y}$, an LML detector with neighborhood size J chooses a symbol vector $\hat{\mathbf{b}}$ that attains the maximum likelihood in its own neighborhood [17]

$$\hat{\mathbf{b}} = \arg \max_{\mathbf{b} \in \Omega_J(\hat{\mathbf{b}})} L(\mathbf{y}|\mathbf{b}), \quad \forall \mathbf{y} \tag{3}$$

where the likelihood function is $L(\mathbf{y}|\mathbf{b}) = -\frac{1}{2}\mathbf{b}^H\mathbf{Hb} + \mathrm{Re}(\mathbf{b}^H\mathbf{Ay})$ with $\mathbf{H} = \mathbf{ARA}$. Since the symbols are equiprobable, the LML detector also achieves the local maximum posterior probability $\hat{\mathbf{b}} = \arg\max_{\mathbf{b} \in \Omega_J(\hat{\mathbf{b}})} p_0(\mathbf{b}|\mathbf{r}, \mathbf{S})$ where the posterior probability is

$$p_0(\mathbf{b}|\mathbf{r}, \mathbf{S}) = p(\mathbf{b})\, p_0(\mathbf{r}|\mathbf{b}, \mathbf{S}) \Big/ \sum\nolimits_{\mathbf{b} \in \Omega_K} p(\mathbf{b})\, p_0(\mathbf{r}|\mathbf{b}, \mathbf{S}).$$

In the special case of $J = K$, the neighborhood of any vector $\Omega_K(\hat{\mathbf{b}}) = \Omega_K$ is the entire set of all vectors. The LML detector in (3) becomes the GML detector that chooses the vector $\hat{\mathbf{b}}$ attaining the global maximum likelihood over the entire Ω_K.

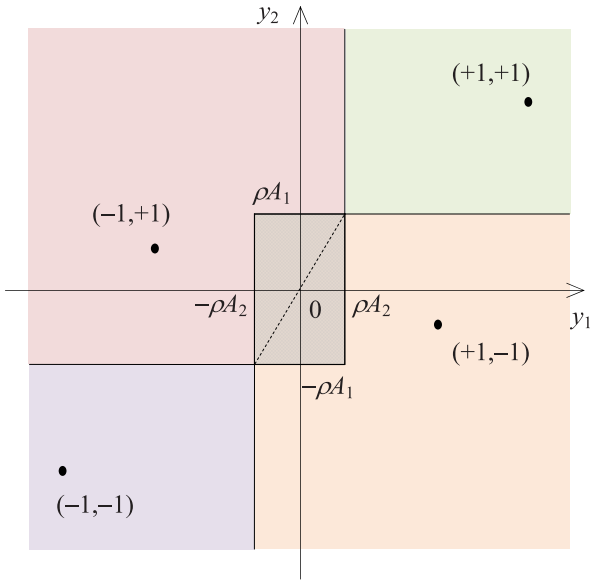


Fig. 1. The LML regions with $J = 1$ and the GML decision regions for the two-bit channel with $R_{12} = \rho > 0$ [18]. The four dots are the noise-free MF-output signals RAb for $\mathbf{b} \in \{-1, +1\}^2$. The color shows the LML region where a bit vector is an LML vector. For example, the LML region of $(+1, -1)$ is orange, which is bounded by the two lines $y_1 = -\rho A_2$ and $y_2 = \rho A_1$. The LML regions for $(-1, +1)$ and $(+1, -1)$ are overlapped in the shaded region where both are LML vectors; however, $(-1, +1)$ is the GML vector above the dashed line, and $(+1, -1)$ is the GML vector below the dashed line. In a non-overlapped region, the only LML vector is also the GML vector. The GML decision regions are the same as the LML regions except for the GML regions of $(-1, +1)$ and $(+1, -1)$ which are separated by the dashed line. The LML and GML detectors perform differently only in the shaded region where an LML detector might not select the GML vector in decision.

While the LML and GML detectors can be applied to any complex symbol constellation, most studies for the performance and computational complexity of the LML and GML detectors in the literature focused on the BPSK signals, i.e., $b_k \in \{-1, 1\}$. In general, the GML detector needs an exhaustive search in Ω_K to obtain the GML vector and therefore is NP-hard in computation. When the number of bits K is finite, an upper bound of GML BER can be obtained but the true BER is hard to obtain [1]. In the large-system limit where the number of bits K and the dimension N of channel vectors tend to infinity with their ratio kept a constant, a formula of the GML BER is obtained by replica analysis [5]–[7].

The LML detector with $J < K$ is determined by (3) where the neighborhood on the right-hand side depends on the vector $\hat{\mathbf{b}}$ to be determined, whereas the right-hand side of the GML detector is independent of the vector to be determined. It can be easily seen [17] that an LML vector with neighborhood size J is an LML vector with neighborhood size M for all $M < J$; therefore, the GML vector is an LML vector with any neighborhood size. In addition to the GML vector, depending on $\mathbf{y}$, there might exist other vectors that are also LML vectors satisfying (3). Fig. 1 shows for a two-bit channel the GML decision regions as well as the LML regions of the LML detector with $J = 1$ where a bit vector is an LML vector. The LML detector with $J < K$ performs worse than the GML detector only in the shaded region where the LML vector might differ from the GML vector. Given $K \geq 3$, as the neighborhood size J increases, the shaded region where multiple LML vectors coexist shrinks and then the BER performance of an LML detector shall improve. In general, an LML detector with a neighborhood size $1 < J < K$ performs better than the LML detector with $J = 1$ and worse than the GML detector.

Based on the definition of the LML detector (3), an LML vector can be obtained by an iterative search [17]. Given an initial vector $\mathbf{b}(0)$, search for the vector $\mathbf{b}(1)$ that attains the maximum likelihood in the neighborhood of $\mathbf{b}(0)$; if $\mathbf{b}(1)$ does not attain the maximum likelihood in its own neighborhood, then search for the vector $\mathbf{b}(2)$ that attains the maximum likelihood in the neighborhood of $\mathbf{b}(1)$; repeat the search until an LML vector $\hat{\mathbf{b}}$ is obtained, which attains the maximum likelihood in its own neighborhood. To avoid the redundant computation, the gradient of likelihood function $L(\mathbf{y}|\mathbf{b})$ can be applied as in the WSLAS detector [18]. In general, the complexity of the LML detector is polynomial in J [17]. In particular, the simplest LML detector with neighborhood size $J = 1$ has a per-bit complexity linear in K [19]. For a small system where K is small, the GML detector significantly outperforms in BER an LML detector with a small neighborhood size [18]. However, as the system size increases, the linear-complexity LML detector even with neighborhood size $J = 1$ can approach the BER of the NP-hard GML detector in a broad range of practical system parameters [19]. In a finite system, a BER upper bound of the LML detector with neighborhood size $J = 1$ is obtained [18], which is comparable to the BER upper bound of the GML detector [1]. In the large-system limit, a formula of the LML BER shall be obtained by the replica analysis in Section III and its relationship with the GML BER [5]–[7] will be studied in Section IV.

C. Local and Global Conditional Mean Detectors

To analyze the performance of the LML detector and its optimality by the replica method, we shall henceforth focus on the BPSK signals transmitted through a real AWGN channel. Specifically, each symbol independently and equiprobably takes on $\{-1, 1\}$ and then $\mathbf{b} \in \Omega_K \equiv \{-1, 1\}^K$. The kth channel vector is written as $\mathbf{s}_k = (s_{1k}, \cdots, s_{Nk})^T/\sqrt{N}$ where s_{ik}'s are i.i.d. with zero mean and unit variance, and therefore $\mathbf{s}_k$ converges to the unit length $\|\mathbf{s}_k\| = 1$ almost surely as $N \to \infty$. The channel noise is $\mathbf{m} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 \mathbf{I}_N)$. Accordingly, the probability density function of $\mathbf{r}$ conditioned on $(\mathbf{b}, \mathbf{S})$ is obtained in (2) by replacing σ_0^2 and N with $2\sigma_0^2$ and $N/2$, respectively. $\mathbf{S}$, $\mathbf{b}$, and $\mathbf{m}$ are assumed to be mutually independent. The prior probability distribution of $\mathbf{b}$ is $p(\mathbf{b}) = 2^{-K}, \forall \mathbf{b} \in \Omega_K$. Without loss of generality, the average symbol energy is assumed to be normalized to unit

$$\frac{1}{K}\sum_{k=1}^{K} A_k^2 = 1. \tag{4}$$

The channel load is $\alpha = K/N$ bits/dimension. In a system where the channel dimension is the product of time and

bandwidth without spatial diversity, the channel load is in the unit of bits/s/Hz. Such a system can be the CDMA system where the channel vectors are code sequences with/without spectral spreading [19].

We assume that $K \to \infty$ and $N \to \infty$ and the channel load $\alpha = K/N$ is fixed in the large-system limit. The empirical distribution of symbol energies A_k^2 with $0 < A_k^2 < A_{\max}^2 < \infty$ converges to a distribution function. The ratio of neighborhood size to the number of bits is denoted by $\beta = J/K$ where the neighborhood size J can also grow with K. As the system size tends to infinity, $\beta \to 0$ for any LML detector with a fixed neighborhood size J and $\beta = 1$ for the GML detector.

The GCM detector well studied in [5]–[7] includes the GML detector as a special case. The BER of the GML detector is obtained through analyzing the GCM detector by the replica method. In order to obtain the BER of the LML detector, we shall analyze the general LCM detector defined below.

Given $(\mathbf{r}, \mathbf{S})$, denote by $\hat{\mathbf{b}}$ the detected vector by the LCM detector. Assuming that the receiver does not know the noise level, a postulated *local* posterior distribution for $\mathbf{b} \in \Omega_J(\hat{\mathbf{b}})$ can be defined by using the controlling parameter σ as

$$p(\mathbf{b}|\mathbf{r}, \mathbf{S}) = \frac{1}{Z_J(\mathbf{r}, \mathbf{S})} p(\mathbf{b}) \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{r} - \mathbf{SAb}\|^2\right) \quad (5)$$

where

$$Z_J(\mathbf{r}, \mathbf{S}) = \sum_{\mathbf{b} \in \Omega_J(\hat{\mathbf{b}})} p(\mathbf{b}) \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{r} - \mathbf{SAb}\|^2\right) \quad (6)$$

is called the *local* partition function. When $J = K$, the local partition function $Z_J(\mathbf{r}, \mathbf{S})$ becomes the global partition function $Z_K(\mathbf{r}, \mathbf{S})$ [5]–[7] where $\Omega_K(\hat{\mathbf{b}}) = \Omega_K$.

The controlling parameter σ is first introduced by Tanaka [5] to define through $Z_K(\mathbf{r}, \mathbf{S})$ the GCM detector that includes the GML and individually optimum detectors as special cases, and to define the general linear detector that includes the decorrelating, linear minimum mean squared error (MMSE), and MF detectors as special cases all by specifying different σ. Actually, we can also define all these detectors in the local sense through $Z_J(\mathbf{r}, \mathbf{S})$ by specifying different σ. Nevertheless, we shall focus on the LCM detector as follows.

Denote by $\langle\cdot\rangle$ the expectation with respect to local posterior distribution $p(\mathbf{b}|\mathbf{r}, \mathbf{S})$. The LCM detector is defined as

$$\hat{b}_k = \operatorname{sgn}\left(\langle b_k \rangle_J\right) = \operatorname{sgn}\left[\frac{1}{Z_J(\mathbf{r}, \mathbf{S})} \times \sum_{\mathbf{b} \in \Omega_J(\hat{\mathbf{b}})} b_k p(\mathbf{b}) \exp\left(-\frac{\|\mathbf{r} - \mathbf{SAb}\|^2}{2\sigma^2}\right)\right], \ \forall k. \quad (7)$$

Since the local posterior distribution depends on $\hat{\mathbf{b}}$ to be determined, the right-hand side of (7) depends on $\hat{\mathbf{b}}$ and therefore the detected LCM vector $\hat{\mathbf{b}}$ is a solution to the fixed-point equation (7).

It is clear that as $\sigma \to 0$, the postulated local posterior distribution (5) eventually puts the unit mass on the maximum-likelihood vector in the neighborhood $\Omega_J(\hat{\mathbf{b}})$ and therefore the LCM detector (7) degenerates to the LML detector (3).

Note that the GCM detector analyzed in [5]–[7] is a special instance of the LCM detector when $J = K$. As $\sigma \to 0$, the GCM detector eventually becomes the GML detector. The GCM detector in the large-system limit has been well analyzed by the replica method in the literature; in particular, a BER formula of the GML detector is obtained [5]–[7].

In the replica method, the BER of a detector is obtained through the correlation between the detected vector $\hat{\mathbf{b}}$ and the transmitted vector denoted by $\mathbf{b}_0$. The correlation between $\hat{\mathbf{b}}$ and $\mathbf{b}_0$ for the kth bit is

$$d_k(\mathbf{r}, \mathbf{S}) = \hat{b}_k b_{0k} \quad (8)$$

and their overall weighted correlation with respect to the energy distribution is

$$d(\mathbf{r}, \mathbf{S}) = \frac{1}{K}\hat{\mathbf{b}}^T \mathbf{A}^2 \mathbf{b}_0 = \frac{1}{K}\sum_{k=1}^{K} A_k^2 d_k(\mathbf{r}, \mathbf{S}). \quad (9)$$

By the self-averaging principle [5] which is assumed to be true in this paper, the correlation for bit k in the large-system limit converges almost surely to $d(A_k) = \lim_{N\to\infty} \langle\langle d_k(\mathbf{r}, \mathbf{S})\rangle\rangle$ where $\langle\langle\cdot\rangle\rangle$ denotes the average over the channel vectors and noise. Then the BER for the kth bit is equal to

$$p_e(A_k) = \Pr(\hat{b}_k b_{0k} = -1) = \lim_{N\to\infty} \left\langle\left\langle (1 - d_k(\mathbf{r}, \mathbf{S}))/2 \right\rangle\right\rangle = (1 - d(A_k))/2. \quad (10)$$

The overall weighted correlation in (9) in the large-system limit can be expressed by (4) and (10) as

$$d = \mathbf{E}_A[A^2 d(A)] = 1 - 2\,\mathbf{E}_A[A^2 p_e(A)]. \quad (11)$$

The expectation $\mathbf{E}_A$ here and henceforth is taken in the distribution function of A_k^2.

III. BER of Local Maximum-Likelihood Detectors

A. LML BER Analysis by Replica Method

We define the *local* free energy as $F_N(\mathbf{r}, \mathbf{S}) = K^{-1} \log Z_J(\mathbf{r}, \mathbf{S})$. To obtain the LML BER by the replica method, the local free energy in the large-system limit shall be analyzed

$$\mathbb{F} = \lim_{N\to\infty} F_N(\mathbf{r}, \mathbf{S}). \quad (12)$$

When $J = K$ (or $\beta = 1$), the local free energy becomes the global free energy. The free energy is an important quantity in statistical mechanics. In the large-system limit the system enters an equilibrium state where the free energy is maximized. A macroscopic quantity like BER is determined at the equilibrium state. Following this idea of replica analysis, in what follows we shall analyze first the saddle point of the local free energy and then obtain the LML BER at the saddle point.

In the replica analysis of the global free energy [5]–[7], four assumptions are adopted. Although the four assumptions are not mathematically justified, they are extensively adopted in broad applications and lead to a number of ground-breaking successes including the analysis of GCM detector [5]–[7]. Hence we extend the four assumptions from analysis of the *global* free energy to analysis of the *local* free energy with necessary modifications. If the four assumptions could

be mathematically justified, the performance of the LML detector obtained from the replica analysis would then be also mathematically rigorous. Nevertheless, even without such a justification the analytical result provides at least a lower bound on the LML performance.

Assumption 1: The local free energy is self-averaging in the large-system limit with respect to the channel vectors and noise.

According to Assumption 1, the local free energy in the large-system limit is equal to

$$\mathbb{F} = \lim_{N\to\infty} \left\langle\!\!\left\langle \frac{1}{\alpha N} \log Z_J(\mathbf{r},\mathbf{S}) \right\rangle\!\!\right\rangle. \tag{13}$$

The GML BER obtained through the replica analysis of the global free energy in the large-system limit [5]–[7] can be considered as the result of analyzing the local free energy in the case of $\beta = 1$. To obtain the LML BER, we shall analyze the local free energy in the case of a fixed J or $\beta \to 0$.

Consider the system of n independent replicas $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \Omega_J(\hat{\mathbf{b}})$ with the joint postulated local posterior distribution

$$p(\mathbf{b}_1, \ldots, \mathbf{b}_n | \mathbf{r}, \mathbf{S}) = Z_J^{-n}(\mathbf{r},\mathbf{S}) \prod_{j=1}^{n} \left[p(\mathbf{b}_j) \exp\left(-\frac{1}{2\sigma^2} \left\| \mathbf{r} - \mathbf{S}\mathbf{A}\mathbf{b}_j \right\|^2 \right) \right] \tag{14}$$

where $Z_J^{-n}(\mathbf{r},\mathbf{S})$ is the reciprocal of

$$Z_J^n(\mathbf{r},\mathbf{S}) = \sum_{\mathbf{b}_1,\ldots,\mathbf{b}_n \in \Omega_J(\hat{\mathbf{b}})} \prod_{j=1}^{n} \times \left[p(\mathbf{b}_j) \exp\left(-\frac{1}{2\sigma^2} \left\| \mathbf{r} - \mathbf{S}\mathbf{A}\mathbf{b}_j \right\|^2 \right) \right]. \tag{15}$$

Then the local free energy can be expressed as

$$\mathbb{F} = \lim_{N\to\infty} \frac{1}{\alpha N} \lim_{n\to 0} \frac{\partial}{\partial n} \log \left\langle\!\!\left\langle Z_J^n(\mathbf{r},\mathbf{S}) \right\rangle\!\!\right\rangle \tag{16}$$

$$= \lim_{n\to 0} \frac{\partial}{\partial n} \lim_{N\to\infty} \frac{1}{\alpha N} \log \left\langle\!\!\left\langle Z_J^n(\mathbf{r},\mathbf{S}) \right\rangle\!\!\right\rangle. \tag{17}$$

Although n is an integer, the following two assumptions have been applied to (16) and (17), respectively.

Assumption 2: The derivative with respect to the integer n in (16) exists at least in the vicinity of $n = 0$.

Assumption 3: The order of the limit in $N \to \infty$ and the limit of the derivative in $n \to 0$ in (17) is exchangeable.

The average of $Z_J^n(\mathbf{r},\mathbf{S})$ over the channel vectors and noise is given by (18), shown at the bottom of this page, where the overhead bar denotes the average over the channel vectors and $\sigma_j = \sigma$ for $j = 1, \cdots, n$.

Define random variables $v_{ij} = K^{-1/2} \sum_{k=1}^{K} A_k s_{ik} b_{jk}$ for $i = 1, \cdots, N$ and $j = 0, 1, \cdots, n$. As $N \to \infty$, v_{ij}'s converge to Gaussian random variables with mean zero. Note that v_{ij}'s and v_{kl}'s for $i \neq k$ are independent and v_{ij}'s for $j = 0, 1, \cdots, n$ and any i have the same covariance matrix $\mathbf{Q}$ such that its jlth element is equal to $Q_{jl} = K^{-1}\mathbf{b}_j^T\mathbf{A}^2\mathbf{b}_l$ for $j, l = 0, 1, \cdots, n$. As far as the BER is considered, the transmitted vector can be assumed to be $\mathbf{b}_0 = [1, \cdots, 1]^T$. It follows from (9) that for $i = 1, \cdots, n$

$$\begin{aligned} Q_{0i} &= K^{-1}\mathbf{b}_0^T\mathbf{A}^2\mathbf{b}_i = K^{-1}\mathbf{b}_0^T\mathbf{A}^2\hat{\mathbf{b}} + K^{-1}\mathbf{b}_0^T\mathbf{A}^2(\mathbf{b}_i - \hat{\mathbf{b}}) \\ &= d(\mathbf{r},\mathbf{S}) + K^{-1}\mathbf{b}_0^T\mathbf{A}^2(\mathbf{b}_i - \hat{\mathbf{b}}). \end{aligned}$$

Since the bit vector $\mathbf{b}_i \in \Omega_J(\hat{\mathbf{b}})$ differs from $\hat{\mathbf{b}}$ by at most J bits,

$$\begin{aligned} K^{-1}\mathbf{b}_0^T\mathbf{A}^2(\mathbf{b}_i - \hat{\mathbf{b}}) &= K^{-1}\sum_{k=1}^{K} A_k^2(b_{ik} - \hat{b}_k) \\ &\geq -2A_{\max}^2 J/K \end{aligned}$$

and

$$K^{-1}\mathbf{b}_0^T\mathbf{A}^2(\mathbf{b}_i - \hat{\mathbf{b}}) \leq 2A_{\max}^2 J/K;$$

but $-1 \leq Q_{0i} \leq 1$ due to (4). Hence,

$$\begin{aligned} Q_{0i} \in D_0 \equiv \big[&\max\{-1, d(\mathbf{r},\mathbf{S}) - 2\beta A_{\max}^2\}, \\ &\min\{1, d(\mathbf{r},\mathbf{S}) + 2\beta A_{\max}^2\}\big]. \end{aligned} \tag{19}$$

Similarly, for $i \neq j$ and $i, j = 1, \cdots, n$

$$\begin{aligned} Q_{ij} &= K^{-1}\mathbf{b}_i^T\mathbf{A}^2\mathbf{b}_j = K^{-1}\sum_{k=1}^{K} A_k^2 b_{ik} b_{jk} \\ &= K^{-1}\sum_{b_{ik}=b_{jk}} A_k^2 - K^{-1}\sum_{b_{ik}\neq b_{jk}} A_k^2 \\ &= 1 - 2K^{-1}\sum_{b_{ik}\neq b_{jk}} A_k^2. \end{aligned}$$

Since $\mathbf{b}_i, \mathbf{b}_j \in \Omega_J(\hat{\mathbf{b}})$ differ by at most $2J$ bits, then

$$2K^{-1}\sum_{b_{ik}\neq b_{jk}} A_k^2 \leq 4A_{\max}^2 J/K$$

and $-1 \leq Q_{ij} \leq 1$ due to (4). Therefore,

$$Q_{ij} \in D_1 \equiv \big[\max\{-1, 1 - 4\beta A_{\max}^2\}, 1\big]. \tag{20}$$

$$\begin{aligned} \left\langle\!\!\left\langle Z_J^n(\mathbf{r},\mathbf{S}) \right\rangle\!\!\right\rangle &= \int_{\mathbb{R}^N} \frac{1}{(2\pi\sigma_0^2)^{N/2}} \overline{\exp\left(-\frac{1}{2\sigma_0^2} \|\mathbf{r} - \mathbf{S}\mathbf{A}\mathbf{b}_0\|^2 \right) \sum_{\mathbf{b}_1,\ldots,\mathbf{b}_n \in \Omega_J(\hat{\mathbf{b}})} \prod_{j=1}^{n} \left[p(\mathbf{b}_j) \exp\left(-\frac{1}{2\sigma^2} \left\| \mathbf{r} - \mathbf{S}\mathbf{A}\mathbf{b}_j \right\|^2 \right) \right]} d\mathbf{r} \\ &= \int_{\mathbb{R}^N} \frac{1}{(2\pi\sigma_0^2)^{N/2}} \sum_{\mathbf{b}_1,\ldots,\mathbf{b}_n \in \Omega_J(\hat{\mathbf{b}})} \left(\prod_{j=1}^{n} p(\mathbf{b}_j) \right) \overline{\prod_{j=0}^{n} \exp\left(-\frac{1}{2\sigma_j^2} \left\| \mathbf{r} - \mathbf{S}\mathbf{A}\mathbf{b}_j \right\|^2 \right)} d\mathbf{r} \\ &= \int_{\mathbb{R}^N} \frac{1}{(2\pi\sigma_0^2)^{N/2}} \sum_{\mathbf{b}_1,\ldots,\mathbf{b}_n \in \Omega_J(\hat{\mathbf{b}})} \left(\prod_{j=1}^{n} p(\mathbf{b}_j) \right) \overline{\prod_{j=0}^{n} \prod_{i=1}^{N} \exp\left[-\frac{1}{2\sigma_j^2} \left(r_i - \frac{1}{\sqrt{N}} \sum_{k=1}^{K} A_k s_{ik} b_{jk} \right)^2 \right]} d\mathbf{r} \end{aligned} \tag{18}$$

The expressions of the ranges D_0 and D_1 in (19) and (20), which Q_{0i} and Q_{ij} belong to, are generally applicable to any LCM detector specified by the relative neighborhood size β. The replica analysis for the GCM detector in [5]–[7] considers only the maximum ranges $Q_{0i} \in D_0 = [-1, 1]$ and $Q_{ij} \in D_1 = [-1, 1]$. This is the special case of (19) and (20) when $\beta = 1$ with the fact of $-1 \le d(\mathbf{r}, \mathbf{S}) \le 1$ and $A_{\max}^2 \ge 1$. In this paper, we have the following conditions that (i) $Q_{0i} \in D_0$ and (ii) $Q_{ij} \in D_1$ for $i \ne j$ and $i, j = 1, \cdots, n$ in addition to $Q_{00} = K^{-1}\mathbf{b}_0^T\mathbf{b}_0 = 1$. When the neighborhood size J is fixed or $\beta \to 0$ as $N \to \infty$. Then D_0 and D_1 in (19) and (20) shrink to the single-point sets $D_0 \to \{d\}$ and $D_1 \to \{1\}$, respectively, in the large-system limit.

The summation over $\mathbf{b}_j$ in (18) can be obtained by first summing over one subshell with $Q_{ij} = K^{-1}\mathbf{b}_i^T\mathbf{A}^2\mathbf{b}_j$ and then integrating over all $\mathbf{Q}$'s that satisfy (i) and (ii). We have

$$\left\langle\left\langle Z^n(\mathbf{r}, \mathbf{S})\right\rangle\right\rangle = \int_{D_0^n}\int_{D_1^{n(n-1)/2}} \exp\left[NG(\mathbf{Q})\right]\mu_N(\mathbf{Q}) \times \prod_{1\le i<j\le n} dQ_{ij}\prod_{i=1}^{n} dQ_{0i} \tag{21}$$

where

$$\mu_N(\mathbf{Q}) = \sum_{\mathbf{b}_1,\ldots,\mathbf{b}_n\in\Omega_K}\prod_{j=1}^{n} p(\mathbf{b}_j)\prod_{0\le i<j\le n}\delta(K^{-1}\mathbf{b}_i^T\mathbf{A}^2\mathbf{b}_j - Q_{ij}) \tag{22}$$

and

$$\exp\left(NG(\mathbf{Q})\right) = \int_{\mathbb{R}^N}\frac{1}{\left(2\pi\sigma_0^2\right)^{N/2}} \times\overline{\prod_{j=0}^{n}\prod_{i=1}^{N}\exp\left[-\frac{1}{2\sigma_j^2}\left(r_i - \sqrt{\alpha}v_{ij}(\mathbf{Q})\right)^2\right]}d\mathbf{r} + O(K^{-1})$$

$$= \left\{\int_{\mathbb{R}}\frac{1}{\sqrt{2\pi\sigma_0^2}}\overline{\prod_{j=0}^{n}\exp\left[-\frac{1}{2\sigma_j^2}\left(r - \sqrt{\alpha}v_j(\mathbf{Q})\right)^2\right]}d\mathbf{r}\right\}^N + O(K^{-1})$$

with the overhead bar denoting the average over v_{ij}'s and then v_j's that have the covariance matrix $\mathbf{Q}$. In (22), the bit vectors are extended to the entire set Ω_K under the condition of (i) and (ii). By Cramer's theorem [20], [21], the probability measure of the empirical mean Q_{ij} as $N \to \infty$ satisfies the large deviation property with a rate function

$$I^{(n)}(\mathbf{Q}) = \sup_{\tilde{\mathbf{Q}}}\left[\sum_{0\le i<j\le n}\tilde{Q}_{ij}Q_{ij} - \mathbf{E}_A\log M(\tilde{\mathbf{Q}})\right]$$

where the moment generating function of $A^2b_ib_j$ is

$$M(\tilde{\mathbf{Q}}) = \sum_{b_1,\cdots,b_n\in\{-1,1\}} 2^{-n}\exp\left(A^2\sum_{0\le i<j\le n}\tilde{Q}_{ij}b_ib_j\right).$$

By Varadhan's theorem [20], [21], we have

$$\lim_{N\to\infty} K^{-1}\log\left\langle\left\langle Z^n(\mathbf{r}, \mathbf{S})\right\rangle\right\rangle = \sup_{\mathbf{Q}}\left[\alpha^{-1}G^{(n)}(\mathbf{Q}) - I^{(n)}(\mathbf{Q})\right] \tag{23}$$

where the supremum is taken under the condition of (i) and (ii). Now the following replica symmetry (RS) assumption is adopted.

Assumption 4: The dominant subshell is invariant under exchange of any two replica indexes i, l with $i, l \ne 0$.

By the assumption of replica symmetry, the macroscopic state in the limit system is determined by a symmetric microscopic state such that $\tilde{Q}_{0i} = E$, $\tilde{Q}_{ij} = F$, $Q_{0i} = m$, and $Q_{ij} = q$ for $i \ne j$ are invariant to replica index. $G^{(n)}(\mathbf{Q})$ and $I^{(n)}(\mathbf{Q})$ become functions of m and q and so can be denoted by $G^{(n)}(m, q)$ and $I^{(n)}(m, q)$. It can be obtained [5]–[7] that

$$G^{(n)}(m, q) = -\frac{n-1}{2}\log\left\{1 + \frac{\alpha}{\sigma^2}(1-q)\right\} - \frac{1}{2}\log\left\{1 + \frac{\alpha(1-q)}{\sigma^2} + \frac{n}{\sigma^2}[\sigma_0^2 + \alpha(1-2m+q)]\right\};$$

and $\sum_{0\le i<j\le n}\tilde{Q}_{ij}Q_{ij} = nEm + 0.5n(n-1)F$, $M(\tilde{\mathbf{Q}}) = \exp(-0.5nA^2F)\int_{\mathbb{R}}\cosh^n(A^2E + \sqrt{A^2F}z)Dz$, therefore

$$I^{(n)}(m, q) = \sup_{E,F}\Big[nEm + 0.5n(n-1)F + 0.5nFq - \mathbf{E}_A\left(\log\int_{\mathbb{R}}\cosh^n[A^2E + \sqrt{A^2F}z]Dz\right)\Big]$$

where $Dz = (2\pi)^{-1/2}\exp(-z^2/2)dz$. We obtain $\lim_{N\to\infty}K^{-1}\log\langle\langle Z^n(\mathbf{r}, \mathbf{S})\rangle\rangle = \alpha^{-1}G^{(n)}(d) - I^{(n)}(d)$ and the local free energy is

$$\mathbb{F} = \lim_{n\to 0}\frac{\partial}{\partial n}\sup_{m\in D_0, q\in D_1}\inf_{E,F} f(m, q, E, F)$$

where

$$f(m, q, E, F) = -\frac{1}{2\alpha}\left\{\begin{array}{l}(n-1)\log\left[1 + \frac{\alpha(1-q)}{\sigma^2}\right] \\ +\log\left[\begin{array}{l}1 + \alpha(1-q)/\sigma^2 \\ +\frac{n}{\sigma^2}\left(\sigma_0^2 + \alpha(1-2m+q)\right)\end{array}\right]\end{array}\right\} + \mathbf{E}_A\left(\log\int_{\mathbb{R}}\cosh^n(A^2E + \sqrt{A^2F}z)Dz\right) - nEm - \frac{n(n-1)}{2}F - \frac{n}{2}F. \tag{24}$$

For the LCM detector, we have to consider the condition of (i) and (ii) when taking the supremum of (23). Therefore, to obtain the LML BER, we consider below only the saddle points that fall inside D_0 and D_1. To this end, take derivative of $f(m, q, E, F)$ with respective to m, q, E and F and let the derivative equal to zero, then we have

$$E = 1/[\sigma^2 + \alpha(1-q)], \tag{25}$$

$$F = [\sigma_0^2 + \alpha(1-2m+q)]/[\sigma^2 + \alpha(1-q)]^2, \tag{26}$$

$$m = \mathbf{E}_A[A^2\int_{\mathbb{R}}\tanh(A^2E + \sqrt{A^2F}z)Dz], \tag{27}$$

$$q = \mathbf{E}_A[A^2\int_{\mathbb{R}}\tanh^2(A^2E + \sqrt{A^2F}z)Dz]. \tag{28}$$

Hence, the local free energy at the saddle point can be expressed as a function

$$\begin{aligned}\mathbb{F} = &-\frac{\sigma_0^2+\alpha(1-2m+q)}{2\alpha\sigma^2+2\alpha^2(1-q)}\\ &-Em\frac{1}{2\alpha}\log\left[1+\frac{\alpha}{\sigma^2}(1-q)\right]-\frac{1}{2}F(1-q)\\ &+\mathbf{E}_A\left(\int_{\mathbb{R}}\log\cosh(A^2E+\sqrt{A^2F}z)Dz\right).\end{aligned} \tag{29}$$

Moreover, the BER $p_e(A)$ is a macroscopic quantity which can also be determined by the saddle point of the local free energy. A further analysis of the BER is presented in the Appendix and we obtain $p_e(A) = Q(AE/\sqrt{F})$ where $Q(x) = (1/\sqrt{2\pi})\int_x^\infty \exp(-t^2/2)dt$. Specificallly, for a bit of energy A^2, if $m \in D_0$ and $q \in D_1$, the BER of the LCM detector in the large-system limit satisfies the fixed-point equation

$$p_e(A) = Q\left(\frac{A}{\sqrt{\sigma_0^2+\alpha(1-2m+q)}}\right). \tag{30}$$

Before deriving the LML BER and its relationship to the GML BER, we first investigate the saddle point solution and the GML BER. If $\sigma \to 0$, the LCM detector (7) degenerates to the LML detector given by (3). The GML detector is a specific case of LML when $\beta \to 1$ so $D_0 = [-1, 1]$ and $D_1 = [-1, 1]$. Hence, we always have $m \in D_0$ and $q \in D_1$. Let $\sigma \to 0$ and then E and F become

$$E = \frac{1}{\alpha(1-q)}, \tag{31}$$

$$F = \frac{\sigma_0^2+\alpha(1-2m+q)}{\alpha^2(1-q)^2}, \tag{32}$$

which yield the saddle point solution of q and m for the GML detector

$$\begin{aligned}q = \mathbf{E}_A\Bigg\{A^2\int_{\mathbb{R}}\tanh^2\Bigg(&\frac{A^2}{\alpha(1-q)}\\ &+\frac{A\sqrt{\sigma_0^2+\alpha(1-2m+q)}}{\alpha(1-q)}z\Bigg)Dz\Bigg\},\end{aligned} \tag{33}$$

$$\begin{aligned}m = \mathbf{E}_A\Bigg\{A^2\int_{\mathbb{R}}\tanh\Bigg(&\frac{A^2}{\alpha(1-q)}\\ &+\frac{A\sqrt{\sigma_0^2+\alpha(1-2m+q)}}{\alpha(1-q)}z\Bigg)Dz\Bigg\}.\end{aligned} \tag{34}$$

In general, $0 \le q \le 1$ and $-1 \le m \le 1$ in (33) and (34); however, $0 \le m \le 1$ is obviously true as BER≤ 0.5 is true. Given any $0 \le m \le 1$, as $q \to 1$, the function $\tanh^2$ in (33) is equal to one for almost every z, and therefore $q = 1$ is a solution to equation (31)-(34). With $q = 1$, (34) leads to

$$m = 1 - 2\,\mathbf{E}_A\left\{A^2Q\left(\frac{A}{\sqrt{\sigma_0^2+2\alpha(1-m)}}\right)\right\}. \tag{35}$$

Note that there might be multiple m's in the solution of (35). If we reduce the size of D_0 and D_1 by decreasing β, then some saddle points that falls outside D_0 or D_1 should be removed.

In this paper, we mainly care about the BER of the LML detector when J is a fixed number. As $\beta \to 0$, D_0 and D_1 shrink to the single-point sets $D_0 \to \{d\}$ and $D_1 \to \{1\}$. Hence, $q = 1$ always falls inside D_1. On the other hand, the overall correlation d is determined by the BER through (11), so it satisfies (35) with m replaced by d. Therefore, we have $m \in D_0$ when $q = 1$. More specifically, we have a saddle point solution $m = d$ and $q = 1$. In this case, the LML BER is

$$p_e(A) = Q\left(\frac{A}{\sqrt{\sigma_0^2+4\alpha\mathbf{E}_A(A^2p_e(A))}}\right). \tag{36}$$

As analyzed above, $q = 1$ is also a solution to the GML saddle-point equation (31)-(34) which also yields the LML BER (36). Hence, the saddle point for the LML detector is one of the solutions to the saddle-point equation for the GML detector with the BER (36). Then we obtain the following proposition.

Proposition 1: Under the four assumptions of the replica method, for a bit of energy A^2 the BER of the LML detector with any fixed neighborhood size J in the large-system limit satisfies the fixed-point equation (36), and the LML BER is always one of the solutions to the GML BER.

While the LML BER (36) is one of the solutions to the GML BER with $q = 1$, it is interesting to know whether or not there is a region of system configuration of channel load and SNR such that the GML detector performs better than the LML detector, i.e., an additional solution of $q < 1$ to the saddle-point equation (33) and (34) exists. In Section IV-A, we will analyze this problem in detail and show the difference between GML and LML.

In general, an LML detector with $\beta \in (0, 1)$ performs in the sense of overall likelihood better than the LML detector with $\beta \to 0$ (or a fixed J) and worse than the GML detector with $\beta = 1$. A mathematically rigorous relationship between the BER and the overall likelihood performance of an LML detector is unknown. Nevertheless, we can conjecture that under the four assumptions of the replica method, the BER of an LML detector with any $\beta \in (0, 1)$ in the large-system limit is upper bounded by the LML BER (36) and lower bounded by the GML BER where the formula of the GML BER [5]–[7] is determined through (31)-(34) with $D_0 = [-1, 1]$ and $D_1 = [-1, 1]$.

B. LML BER Analysis by Gaussian Approximation

The same BER formula (36) shall be derived for the LML detector with a fixed neighborhood size $J = 1$ by using the method of Gaussian approximation. According to (3), in the signal space of $\mathbf{y}$ the LML region where a bit vector b is an LML point can be defined by $V(\mathbf{b}) = \left\{\mathbf{y} \in \mathbb{R}^K \,|L(\mathbf{y}|\mathbf{b}) \ge L(\mathbf{y}|\mathbf{a}), \forall \mathbf{a} \in \Omega_1(\mathbf{b})\right\}$. The LML regions of different bit vectors might be mutually overlapped in certain regions where multiple LML points coexist. Nevertheless, the LML region can be considered as the decision region of the LML detector. Unlike the GML detector that achieves the unique minimum probability of error, due to overlapping of

multiple LML regions, there exist a number of LML detectors each associated with a probability of error $P_e \equiv \Pr(\hat{\mathbf{b}} \neq \mathbf{b})$. It can be obtained [17], [18] that the LML region $\mathbf{V}(\mathbf{b})$ is a convex hull consisting of K hyperplanes each of which is perpendicular to an axis

$$\mathbf{b} \otimes (\mathbf{y} - (\mathbf{R} - \mathbf{I})\mathbf{A}\mathbf{b}) \geq \mathbf{0} \tag{37}$$

where the multiplication $\otimes$ and inequality $>$ are element-wise.

The LML BER formula (36) shall be obtained from (37) by the method of Gaussian approximation. The BER of bit k is upper bounded by the sum of the probabilities that $\mathbf{b}$ is transmitted but $\mathbf{y}$ is inside the LML region of an error vector $\hat{\mathbf{b}}$ with $\hat{b}_k \neq b_k$. We are interested only in the separation of $\hat{b}_k \neq b_k$ by the boundary hyperplanes of the LML regions. In terms of (37), $\hat{b}_k$ and b_k for $\hat{b}_k \neq b_k$ are separated by 2^{K-1} boundary hyperplanes

$$\hat{b}_k \left(y_k - \sum_{j=1}^{K} R_{kj} A_j \hat{b}_j \right) \geq -A_k \tag{38}$$

each of which is determined by a combination of $K-1$ bits b_i for $i \neq k$. For example, in Fig. 1 the two lines $y_1 = -\rho A_2$ and $y_1 = \rho A_2$ that correspond to $b_2 = -1$ and $b_2 = 1$ respectively separate $b_1 = -1$ and $b_1 = 1$. Then (38) is used to obtain an approximated BER of bit k.

Inequality (38) is equivalent to a detector that decides $\hat{\mathbf{b}}$ as the transmitted $\mathbf{b}$ while the other bits are detected by (38) with exchanging k and i. When deciding a bit, the detector needs to know the decisions of all others. Let $\boldsymbol{\varepsilon} = (\mathbf{b} - \hat{\mathbf{b}})/2$ be the error vector. It follows from (38) that the BER of bit k is

$$\begin{aligned} p_e(A_k) &= \Pr\left[\hat{b}_k \left(n_k + 2 \sum_{j=1}^{K} R_{kj} A_j \varepsilon_j \right) \geq -A_k \middle| \varepsilon_k \neq 0 \right] \\ &= \Pr\left(n_k + 2 \sum_{j=1, j\neq k}^{K} R_{kj} A_j \varepsilon_j \geq (2R_{kk} - 1) A_k \right) \end{aligned} \tag{39}$$

where the second equality is due to $\hat{b}_k \varepsilon_k = -1$ and the symmetry of noise and interference. As $N \to \infty$, R_{kk} converges almost surely to one.

The term of noise and interference $w_k = n_k + 2\sum_{j=1, j\neq k}^{K} R_{kj} A_j \varepsilon_j$ in (39) has zero mean. The probability of the error ε_i for $i \neq k$ is $\Pr(\varepsilon_i \neq 0) = p_e(A_i)$. If the crosscorrelations of bit errors are ignored, the variance of w_k is equal to

$$\sigma_k^2 = \sigma_0^2 + 4 \sum_{j=1, j\neq k}^{K} R_{kj}^2 A_j^2 p_e(A_j). \tag{40}$$

As $N \to \infty$, NR_{kj}^2 for $j \neq k$ converges almost surely to one and then the variance of w_k converges almost surely to $\sigma_0^2 + 4\alpha \mathbf{E}_A(A^2 p_e(A))$. Hence, approximating w_k by a Gaussian random variable, we obtain the same BER formula (36).

C. Solving Numerically the LML BER

According to (35), the overall weighted correlation d for the LML detector satisfies the fixed point equation

$$d = 1 - 2\mathbf{E}_A \left(A^2 Q \left(\frac{A}{\sqrt{\sigma_0^2 + 2\alpha(1-d)}} \right) \right) \tag{41}$$

Once solving d from (41), by means of (30) the BER for a bit of energy A^2 is equal to

$$p_e(A) = Q\left(\frac{A}{\sqrt{\sigma_0^2 + 2\alpha(1-d)}} \right). \tag{42}$$

For the equal energy system, (41) and (42) become $d = 1 - 2Q\left(1 \Big/ \sqrt{\sigma_0^2 + 2\alpha(1-d)}\right)$, and $p_e = Q\left(1 \Big/ \sqrt{\sigma_0^2 + 2\alpha(1-d)}\right)$, respectively. For the uncoded system, the energy per bit E_b divided by the noise power spectral density N_0 is determined by $E_b/N_0 = 1/(2\sigma_0^2)$.

In general, there might be up to three solutions to (41). Consider a solution d^*. If the derivative of the right-hand side of (41) at d^* is less than one, then d^* can be simply obtained numerically by the Picard iteration of (41) with an initial d_0 in the neighborhood of d^*, that is

$$d_{k+1} = 1 - 2\mathbf{E}_A \left(A^2 Q \left(\frac{A}{\sqrt{\sigma_0^2 + 2\alpha(1-d_k)}} \right) \right). \tag{43}$$

On the other hand, if the derivative of the right-hand side of (41) at d^* is greater than one, then the iteration shall change d in the opposite direction indicated by the sign of the difference between the left and right-hand sides of (41), that is

$$d_{k+1} = d_k + \gamma \left[d_k - 1 + 2\mathrm{E}_A \left(A^2 Q \left(\frac{A}{\sqrt{\sigma_0^2 + 2\alpha(1-d_k)}} \right) \right) \right], \tag{44}$$

where γ is a small positive number.

Consider that K bits are divided into M groups according to their energies. The i-th group has K_i bits, takes energy A_i^2, and its percentage of population is $K_i/K \to \lambda_i$ for $i = 1, \ldots, M$ as $K \to \infty$. Then (43) and (44) become

$$d_{k+1} = 1 - 2\sum_{i=1}^{M} \lambda_i A_i^2 Q \left(\frac{A_i}{\sqrt{\sigma_0^2 + 2\alpha(1-d_k)}} \right), \tag{45}$$

and

$$d_{k+1} = d_k + \gamma \left[d_k - 1 + 2\sum_{i=1}^{M} \lambda_i A_i^2 Q \left(\frac{A_i}{\sqrt{\sigma_0^2 + 2\alpha(1-d_k)}} \right) \right], \tag{46}$$

respectively.

IV. Optimality of the Local Maximum-Likelihood Decectors

A. Performances of LML and GML Detectors in the Equal-Energy Distribution

It is interesting to know whether or not there is a region of system configuration of channel load and SNR such that the GML detector performs better than the LML detector, i.e., an additional solution of $q < 1$ to the saddle-point equation (33) and (34) exists. To investigate the possible performance difference between the LML and GML detectors, we consider the equal-energy distribution so that the saddle-point equation of (q, m) for the global free energy in (33) and (34) becomes

$$q = \int_{\mathbb{R}} \tanh^2 \left(\frac{1}{\alpha(1-q)} + \frac{\sqrt{\sigma_0^2 + \alpha(1-2m+q)}}{\alpha(1-q)} z \right) Dz, \tag{47}$$

$$m = \int_{\mathbb{R}} \tanh \left(\frac{1}{\alpha(1-q)} + \frac{\sqrt{\sigma_0^2 + \alpha(1-2m+q)}}{\alpha(1-q)} z \right) Dz. \tag{48}$$

To seek the possible solution of $q < 1$, we investigate numerically the following relative error function of $1-q$ and $1-m$

$$f(q,m) = \frac{(q^*(q,m)-q)^2 + (m^*(q,m)-m)^2}{(1-q)^2 + (1-m)^2} \tag{49}$$

where $q^*(q,m)$ and $m^*(q,m)$ are functions of q and m evaluated by the right-hand sides of (47) and (48), respectively. If $f(q,m) < 10^{-7}$, then the pair of values (q,m) is considered to numerically reach a saddle point that solves the saddle-point equations of (47) and (48). Prior to doing so, we cite the result in [5] that the de Almeida-Thouless (AT) stability line specifies the local stability of the replica symmetry saddle-point solution against replica symmetry breaking perturbation. The AT line is given by $1 - \alpha E^2 \int_{\mathbb{R}} \mathrm{sech}^4(E + \sqrt{F}z) Dz = 0$. The replica symmetry saddle-point solution is valid if the left-hand side is positive.

As observed for the GML BER [5], the LML BER becomes nonmonotonically dependent on the channel load α as α becomes larger. Under some conditions multiple BERs coexist. As pointed out in [5], the coexistence of multiple BERs "is essentially the same as the coexistence of phases in physical systems (such as water and ice at 0°C), and such a phenomenon is called phase coexistence in statistical mechanics."

By the numerical investigation of $f(q,m)$, two regions of system parameters $(\alpha, E_b/N_0)$ are found: region A where all saddle-point solutions have $q > 0.999$, and region B where in addition to the saddle-point solutions of $q > 0.999$ there exists an additional saddle-point solution such that $q < 0.7$. In region A, since $q > 0.999$ is very close to one, even if a saddle-point solution of GML has $q \neq 1$, the value m that determines the GML BER is close to the m that determines LML BER that has $q = 1$ and therefore the LML and GML BERs are numerically indistinguishable. Thus, we can say that in region A the LML and GML detectors perform identically at least in the numerical sense. In region B, the saddle-point solution of GML with $q < 0.7$ yields the GML BER apparently different from the LML BER. Hence, we can say that in region B the LML detector performs worse than the GML detector only for the additional bad solution (i.e., the high-BER solution as termed in [5]) with $q < 0.7$.

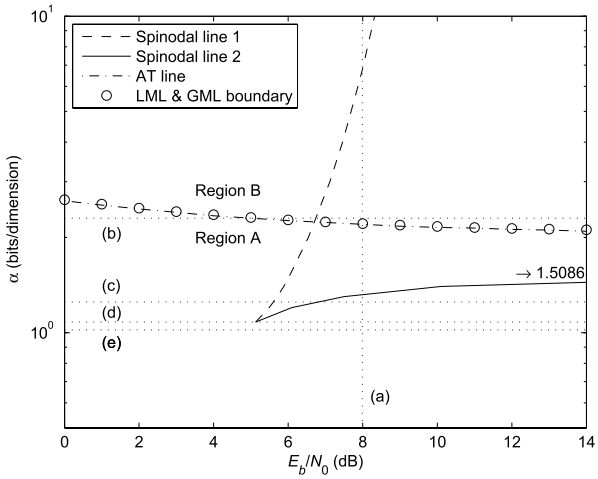


Fig. 2. A practical region of α and E_b/N_0. Lines (a) – (e) indicate α and E_b/N_0 used in Fig. 3 (a) – (e), respectively.

It is interesting that the boundary that separates regions A and B is numerically overlapped with the AT line as shown in Fig. 2. Region A is below the AT line where the replica symmetry saddle-point solution is invalid and region B is above the AT line where the replica-symmetry saddle-point solution is valid. The good solution that results in a low BER exists below spinodal line 1, and the bad solution that results in a high BER exists above spinodal line 2. As shown in Fig. 3 (a) – (e), in region A the saddle-point solutions for the GML and LML detectors are identical. However, in region B the GML detector has two bad solutions. One of the solution has a lower BER with $q < 0.7$ and the other with $q > 0.999$. The saddle-point solution with $q < 0.7$ is approached by the bit vectors with a crosscorrelation much less than one that are not located in a neighborhood of a finite size, and therefore cannot be obtained by the LML detector. The saddle-point solution with $q > 0.999$ is approached by the bit vectors that are located approximately (in the numerical sense) in the same neighborhood and therefore can be approached by the LML detector. The LML and GML detectors have the same good solution and the intermediate solution between the bad and good solutions. In other words, in region B the GML detector performs differently from the LML detector only for the additional bad solution that the LML detector cannot achieve.

On the basis of numerical results, we conjecture that in the equal-energy distribution the AT line separates the two regions of channel load and SNR. In the region where the replica symmetry saddle-point solution is invalid, the LML and GML detectors perform identically with the only saddle-point solution of $q = 1$. In the region where the replica symmetry saddle-point solution is valid, the LML and GML detectors still perform identically with the saddle-point solution of $q = 1$ but the GML detector has an additional bad saddle-point solution of $q \neq 1$ that the LML detector cannot achieve.

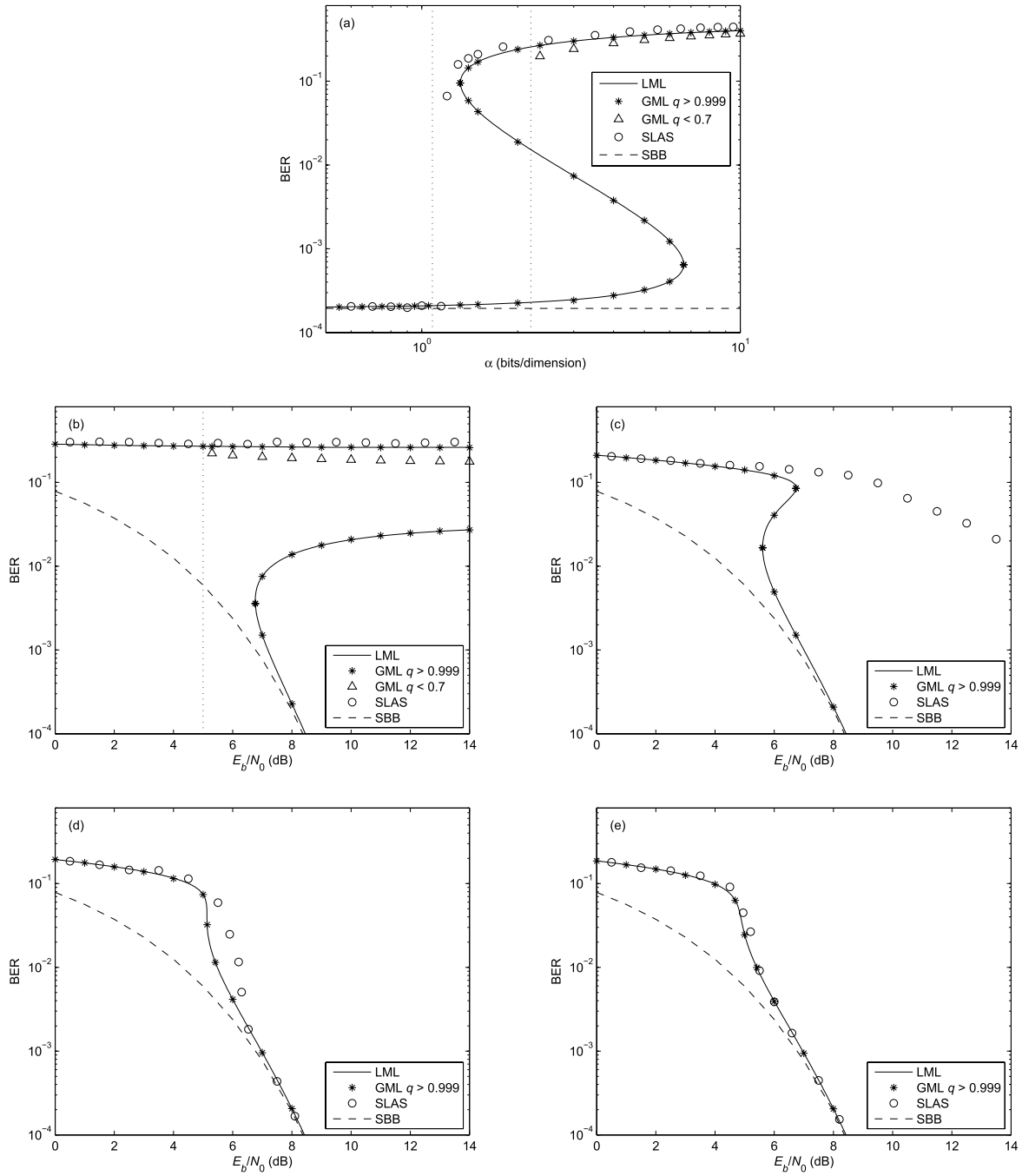


Fig. 3. The LML BER is obtained by the formulas (36). To determine the GML BER, the minimum point of $f(q, m)$ is numerically chosen to be the initial point to iterate (33) and (34) until $f(q, m) < 10^{-7}$. The SLAS BER is obtained by the SLAS detector [19] in simulation with a random initial bit vector. The equal-energy distribution is considered in all cases. The single-bit bound (SBB) is also shown. (a) BER versus α with $E_b/N_0 = 7.99$ dB. The left vertical line indicates $\alpha = 1.08$ bits/dimension and the right vertical line indicates $\alpha = 2.20$ bits/dimension which separates regions A and B. (b) BER versus E_b/N_0 with $\alpha = 2.30$ bits/dimension. The vertical line indicates $E_b/N_0 = 5.0$ dB that separates regions A and B. (c) BER versus E_b/N_0 with $\alpha = 1.25$ bits/dimension. (d) BER versus E_b/N_0 with $\alpha = 1.08$ bits/dimension. (e) BER versus E_b/N_0 with $\alpha = 1.02$ bits/dimension. The values of $(\alpha, E_b/N_0)$ in (c) – (e) all are in region A. In these figures, the gap between the LML and GML BERs is small and so is the gap between their multiuser efficiencies.

B. Achievable Channel Load

The multiuser efficiency [1] of a detector is defined as $\eta = Q^{-2}(p_e(A))\,\sigma_0^2/A^2$ where Q^{-2} denotes the squared inverse of Q function. By means of (36), the multiuser efficiency of the LML detector is the solution to the fixed-point equation

$$\eta = \frac{\sigma_0^2}{\sigma_0^2 + 4\alpha \mathbf{E}_A(A^2 p_e(A))}. \tag{50}$$

η measures the efficiency of power usage compared with the single-bit channel. The effective signal-interference ratio (SIR) for a bit of energy A^2 is $\mathrm{SIR}(A) = \eta A^2/\sigma_0^2$. The asymptotic multiuser efficiency (AME) [1] as the noise vanishes is

$$\hat{\eta} = \lim_{\sigma_0 \to 0} \frac{\sigma_0^2}{\sigma_0^2 + 4\alpha \mathbf{E}_A(A^2 p_e(A))}. \tag{51}$$

Since $p_e(A)$ can be multiple-valued, so are the multiuser efficiency and asymptotic multiuser efficiency.

In the large-system limit the BER for the bit of energy A^2 is $p_e(A)$. If there is no interference bit, the BER reaches the SBB $p_s(A) = Q(A/\sigma_0)$ which converges exponentially fast to zero as $\sigma_0 \to 0$. If the AME equals one, in the high SNR regime the LML BER $p_e(A)$ can approach $p_s(A)$ as if there was no interference bit. Then we define the cutoff channel load (CCL) in the high SNR regime as

$$\alpha^* = \sup_{\alpha>0}\{\alpha : \hat{\eta} = 1\}.$$

The CCL α^* is the marginal channel load at which the bad solution disappears in the high SNR regime. The CCL α^* divides the channel load into two regions. All channel loads $\alpha < \alpha^*$ are asymptotically achievable, under which all bits can achieve the SBB BER in the high SNR regime. On the other hand, all channel loads $\alpha > \alpha^*$ are not achievable.

To obtain the CCL for the LML detector, we shall analyze the LML BER in the high SNR regime. First, we have the following lemma which is proved in the Appendix.

Lemma 1: For the fixed-point equation of p

$$p = Q\left(1/\sqrt{4\alpha p}\right), \quad \alpha > 0 \tag{52}$$

there exist $\alpha_0 \cong 1.5086$ and $p_0 \cong 0.1169$ such that (i) if $\alpha < \alpha_0$, only $p = 0$ solves (52); (ii) if $\alpha = \alpha_0$, then both $p = p_0$ and $p = 0$ solves (52); (iii) if $\alpha > \alpha_0$, in addition to $p = 0$ there exist two points $p_1 \in (0, p_0)$ and $p_2 \in (p_0, 1)$ that solve (52).

In terms of Lemma 1, the LML CCL in the equal-energy distribution is $\alpha^* = \alpha_0 \cong 1.5086$ bits/dimension. As shown in Fig. 2, α_0 is the marginal channel load such that the bad solution disappears in the high SNR regime. Consequently, the LML and GML detectors have the same CCL. It is interesting that the LML/GML CCL α_0 is greater than 1 bit/dimension. In contrast, the channel load equals 1 bit/dimension for a set of orthogonal channel vectors and for the TDMA/FDMA systems where both transmitter and receiver have a single antenna. Moreover, the AMEs of the MMSE, decorrelating and MF detectors all are smaller than one [3] and therefore their CCLs are equal to zero.

The LML CCL is low in the equal-energy distribution and can be arbitrarily high in an unequal-energy distribution. To obtain this, we extend the result of LML CCL to an arbitrary energy distribution.

Proposition 2: The LML CCL in an arbitrary energy distribution is equal to

$$\alpha^* = \frac{I}{4\mathbf{E}_A\left(A^2 Q\left(A/\sqrt{I}\right)\right)} \tag{53}$$

where the interference energy at α^* satisfies

$$I = \frac{\mathbf{E}_A^2\left(A^3 \exp\left(-A^2/(2I)\right)\right)}{8\pi\,\mathbf{E}_A^2\left(A^2 Q\left(A/\sqrt{I}\right)\right)}. \tag{54}$$

To obtain α^* numerically in an arbitrary energy distribution, we can first iteratively solve I from (54) and then obtain α^* from (53). Letting $A = 1$, (53) and (54) degenerate to (61) and (62) in the Appendix that iteratively determine the LML CCL in the equal-energy distribution.

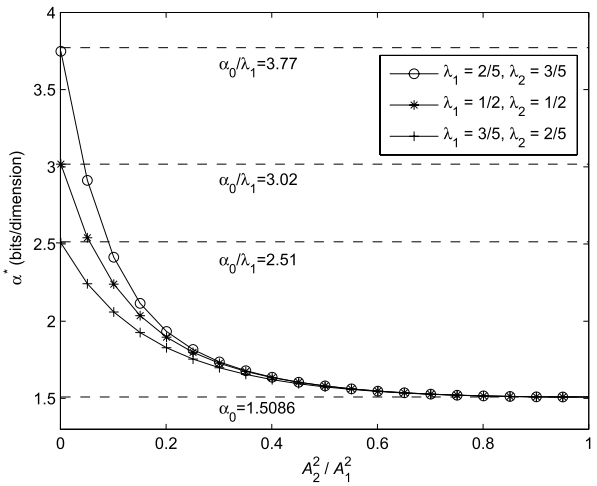


Fig. 4. The cutoff channel load α^* versus energy distribution in the two-class system where the ith class has energy A_i^2 and percentage of population $\lambda_i, i = 1, 2$.

As shown in Fig. 4 for the two-class system, an unequal energy distribution can significantly increase the region of achievable channel load. As the energy difference between the two classes increases, the CCL monotonically increases and can be arbitrarily close to α_0/λ_1 if A_2 is sufficiently small. This is true in general, that is, without limitation on the energy distribution any channel load $\alpha > 0$ is achievable for the LML detector. Moreover, $\alpha^* \geq \alpha_0$ where the equality holds for and only for the equal-energy distribution. Hence, an unequal-energy distribution is more preferable than the equal-energy distribution.

Remark: For the LML detector, the total effective multiuser interference is $4\alpha\mathbf{E}_A(A^2 p_e(A))$. A bit of energy A^2 interferes others only through $4\alpha A^2 p_e(A)$. Below the spinodal line 2 in Fig. 2 where the bad solution disappears, the BER $p_e(A)$ decreases exponentially fast to zero with the energy increasing, and therefore a strong bit yields a weak interference. In particular, a bit with $A \to \infty$ does not interfere others since $A^2 p_e(A) \to 0$ and thus the LML detector is effective in near-far resistance [1]. Moreover, a weak bit also yields a weak interference. Consequently, the overall interference is low. This is why the LML detector is superior to all linear detectors. For example, the linear MMSE detector has the multiuser efficiency $\eta = \sigma_0^2/[\sigma_0^2 + \alpha\mathbf{E}_A(A^2/(1 + \eta A^2\sigma_0^{-2}))]$ [3]. A bit of energy A^2 interferes others through $A^2/(1 + \eta A^2\sigma_0^{-2})$, which does not vanish as energy increases and therefore the overall interference is high.

C. Practical Achievability of GML BER

As shown in Fig. 2, when the channel load α is lower than the spinodal line 2 where the bad solution disappears, there is only a good solution and the LML and GML detectors perform identically. Specifically, when $\alpha < 1.08$ bits/dimension, there is no other solution but a good solution for all SNR. If $E_b/N_0 > 5.13$ dB, the channel load can be higher than

1.08 bits/dimension. As SNR tends to infinity, the highest channel load is $\alpha = 1.5086$ bits/dimension. These facts imply the possible achievability of the NP-hard GML BER in a practical large system by a linear-complex LML detector that uses only the local information.

To investigate this the SLAS detector [19] is examined in a simulation. The SLAS detector is an LML detector with neighborhood size of $J = 1$ and has a linear per-bit computational complexity [19]. The simulation result for equal-energy system is reported here. In a trial, a transmission of $K = 8192$ bits is considered. A random bit vector is used for the SLAS detector to initiate the likelihood ascent search. The BER is obtained by averaging over a number of transmissions such that at least a total of 300 bit errors occur. Fig. 3 (a) shows BER versus α with $E_b/N_0 = 7.99$ dB. The SLAS detector behaves differently in three regimes of channel load. In the regime of $\alpha < 1.15$ bits/dimension where there is only the good LML/GML solution, the SLAS detector approaches the BER that is close to the SBB. In the regime between the spinodal line 2 and the AT line with $1.15 < \alpha < 2.20$ bits/dimension where one good and one bad LML/GML solution coexist, the SLAS detector can only approach the bad BER as predicted by Tanaka [5]. These two regimes are in region A where the replica symmetric saddle-point solution is invalid. On the other hand, in the regime above the AT line (region B) with $\alpha > 2.20$ bits/dimension where the good LML/GML solution, the bad LML/GML solution, and the bad GML solution coexist, the SLAS detector can only approach the higher LML/GML BER of the bad solution. This confirms the preceding analysis that the lower GML BER of the bad solution with $q < 0.7$ cannot be approached by an LML detector. Fig. 3 (b) shows BER versus E_b/N_0 with $\alpha = 2.3$ bits/dimension. The left side of the vertical line with $E_b/N_0 = 5.0$ dB is in region A where there is only one bad LML/GML solution and the right side is in region B where an additional bad GML solution appears. However, as expected, in the entire considered region the SLAS detector can only approach the higher bad LML/GML solution. Fig. 3 (c) shows BER versus E_b/N_0 with $\alpha = 1.25$ bits/dimension. The system parameters $(\alpha, E_b/N_0)$ are in region A below the AT line, and therefore the GML and LML detectors perform identically. For $E_b/N_0 < 6.73$ dB above the spinodal line 2, the bad LML/GML solution exists and the SLAS detector can only approach the bad solution. For $E_b/N_0 > 6.73$ dB below the spinodal line 2, only the good LML/GML solution exists. However, since $\alpha = 1.25$ bits/dimension is too close to the spinodal line 2 as shown in Fig. 2, the SLAS detector approaches a BER with a big gap to the analytical solution. The reason is that when the system parameters $(\alpha, E_b/N_0)$ are close to the spinodal line 2, the error function $f(q, m)$ is small and the SLAS performance is sensitive to the system size. It is observed that the BER gap decreases as the number of bits increases. That is, the number of bits $K = 8192$ is not sufficiently large for the SLAS detector to approach the analytical solution. As shown in Fig. 3 (d) with $\alpha = 1.08$ bits/dimension that is farther away from the spinodal line 2, the SLAS detector can approach the good LML/GML BER except slightly affected by the spinodal line 2 when E_b/N_0 is close to 5.13 dB at which the two spinodal lines intersect. As shown in Fig. 3 (e) with $\alpha = 1.02$ bits/dimension where the system parameters are far away from the spinodal line 2, the SLAS detector can approach the only and good LML/GML BER. These simulation results demonstrate that the linear-complex SLAS detector using only the local information can approach the NP-hard GML performance in a practical large system. This is theoretically significant and might be useful in practice.

The numerical results in Section IV.B demonstrate that the BER difference between the LML with $\beta \to 0$ and the GML with $\beta = 1$ is small. Since the BER of any LML detector with any β shall be upper bounded by the LML BER with $\beta \to 0$ and lower bounded by the GML BER with $\beta = 1$, the BER difference among all LML detectors with any $\beta \in [0, 1]$ in the large system limit is small. This implies that in a sufficiently-large practical system the BER difference among all LML detectors with any neighborhood size J is small. The above simulation result demonstrates that in a system of $K = 8192$ bits the SLAS detector, which is an LML detector with the smallest neighborhood size of $J = 1$, can already approach the BER with a small gap to the analytical GML BER. This means that any other LML detector with a neighborhood size $J > 1$ should not considerably improve the BER. On the other hand, the computational complexity of an LML detector with a neighborhood size J grows polynomially fast with J [17]. Hence, as a guideline for the application of the LML detector in a practical sufficiently-large MIMO system, it is preferred to employ the LML detector with the smallest neighborhood size of $J = 1$ in order to achieve a better tradeoff between the BER performance and the computational complexity.

V. Conclusions

We have extended the replica analysis of the GML detector to the LML detector in the large-system limit. By the replica method a BER formula of the LML detector with a fixed neighborhood size is obtained. The same LML BER formula is obtained by the conventional method of Gaussian approximation for the LML detector with the unit neighborhood size. The LML BER is always one of the solutions to the GML BER. Based on the numerical result, we conjecture that in the equal-energy distribution the LML and GML detectors perform identically below the AT stability line; above the AT line the GML detector performs differently from the LML detector only for an additional bad BER solution which the LML detector cannot attain. In the high SNR regime, the LML and GML detectors both can achieve the AWGN BER when the channel load is up to 1.5086 bits/dimension in an equal-energy distribution. The achievable channel load can be arbitrarily higher in an unequal-energy distribution. In the large-system limit the BER difference between the LML detector with the smallest neighborhood size and the GML detector with the largest neighborhood size is numerically small. Therefore, the BER difference among all LML detectors with any neighborhood size must be small, and so be the difference of their multiuser efficiencies. In the regime below the spinodal line 2, the linear-complexity SLAS detector that

is an LML detector with the unit neighborhood size and uses only the local information can approach the NP-hard GML BER predicted by the analysis. In a practical sufficiently-large MIMO system it is preferred to employ an LML detector with the unit neighborhood size to achieve a better tradeoff between the BER performance and the computational complexity. While the replica analysis of the LML detector for binary symbols is successful, the corresponding analysis for the general constellation remains open.

APPENDIX

Proof of (30)

Consider user 1 and its LCM BER is determined from the correlation by $p_e(A_1) = (1 - d(A_1))/2$. The correlation of the LCM detector output from (7) and (8) is

$$d(A_1) = \lim_{N\to\infty} \left\langle\!\left\langle \operatorname{sgn}\left(\langle b_1\rangle_J\right)\right\rangle\!\right\rangle. \tag{55}$$

To evaluate $d(A_1)$, consider a replica of the modified system with the local posterior distribution

$$p(\mathbf{b}_1,\cdots,\mathbf{b}_n|h;\mathbf{r},\mathbf{S}) = Z_J^{(-n)}(h;\mathbf{r},\mathbf{S}) \exp\left(h\prod_{j=1}^{i} b_{l_j 1}\right) \times p(\mathbf{b}_1,\cdots,\mathbf{b}_n|\mathbf{r},\mathbf{S})$$

where $Z_J^{(-n)}(h;\mathbf{r},\mathbf{S})$ is the reciprocal of

$$Z_J^{(n)}(h;\mathbf{r},\mathbf{S}) = \sum_{\mathbf{a}\in\Phi(\mathbf{r},\mathbf{S})} \sum_{\mathbf{b}_1,\cdots,\mathbf{b}_n\in\Omega_J(\mathbf{a})} \exp\left(h\prod_{j=1}^{i} b_{l_j 1}\right) \times p(\mathbf{b}_1,\cdots,\mathbf{b}_n|\mathbf{r},\mathbf{S})$$

and l_j's are arbitrary i distinct indexes out of $1,2,\cdots,n$. It is clear that

$$\lim_{n\to 0}\frac{\partial}{\partial h}\left\langle\!\left\langle \log Z_J^{(n)}(h;\mathbf{r},\mathbf{S})\right\rangle\!\right\rangle\Big|_{h=0} = \left\langle\!\left\langle \langle b_1\rangle^i\right\rangle\!\right\rangle.$$

Due to the similarity between $Z_J^{(n)}(h;\mathbf{r},\mathbf{S})$ and $Z_J^n(\mathbf{r},\mathbf{S})$ and by following the similar lines in Section III.A, we obtain

$$\left\langle\!\left\langle Z_J^{(n)}(h;\mathbf{r},\mathbf{S})\right\rangle\!\right\rangle = \int_{D_0^n}\int_{D_1^{n(n-1)/2}} \exp\left(NG(\mathbf{Q})\right)\mu_N(h;\mathbf{Q}) \times \prod_{1\le j<l\le n} dQ_{jl}\prod_{j=1}^{n} dQ_{0j} \tag{56}$$

where

$$\mu_N(h;\mathbf{Q}) = \sum_{\mathbf{b}_1,\cdots,\mathbf{b}_n\in\Omega_K} \exp\left(h\prod_{j=1}^{i} b_{l_j 1}\right)\prod_{j=1}^{n} p(\mathbf{b}_j) \times \prod_{0\le j<l\le n}\delta(K^{-1}\mathbf{b}_j^T\mathbf{A}^2\mathbf{b}_l - Q_{jl}).$$

Note that in the summation over $\mathbf{b}_1,\cdots,\mathbf{b}_n\in\Omega_K$, only those terms with b_{j1} for $j=1,\cdots,n$ are relevant to h. The moment generating function related to h is

$$M(h;\tilde{\mathbf{Q}}) = \sum_{b_1,\ldots,b_n\in\{-1,1\}} \frac{1}{2^n} \times \exp\left(A_1^2\sum_{0\le j<l\le n}\tilde{Q}_{jl}b_jb_l + h\prod_{j=1}^{i} b_{l_j}\right).$$

Thus, by exchanging the order of $N\to\infty$ and $n\to 0$ and under the replica symmetry, we obtain

$$\begin{aligned}\lim_{N\to\infty}\left\langle\!\left\langle\langle b_1\rangle^i\right\rangle\!\right\rangle &= \lim_{h\to 0}\lim_{n\to 0}\frac{\partial}{\partial h}\lim_{N\to\infty}\left\langle\!\left\langle \log Z_J^{(n)}(h;\mathbf{r},\mathbf{S})\right\rangle\!\right\rangle \\ &= \lim_{n\to 0}\frac{\partial}{\partial h}\mathbf{E}_A[\log M(h;\tilde{\mathbf{Q}})]\Big|_{h=0} \\ &= \int_{\mathbb{R}}\tanh^i\left(A_1^2E + \sqrt{A_1^2F}z\right)Dz. \end{aligned}\tag{57}$$

The function $\operatorname{sgn}(\langle b_1\rangle)$ can be approached by a series of polynomials of $\langle b_1\rangle^i$. Therefore, the BER for a bit of energy A^2 is $p_e(A) = Q(AE/\sqrt{F})$, which produces (30). (Q.E.D.)

Proof of Lemma 1

It is clear that $p=0$ always solve (52).

Now we show that α_0 exists. Consider the function $g(p) = p - Q\left(1/\sqrt{4\alpha p}\right)$. Let the first order of derivative of $g(p)$ with respect to p

$$g'(p) = 1 - \frac{1}{4\sqrt{2\pi\alpha}\,p^{3/2}\exp\left(1/(8\alpha p)\right)} \tag{58}$$

be greater than or equal to zero. Considering only the quadratic term in the Tailor expansion of the exponential function, we obtain that if $\alpha \le \pi^{1/3}/8$, then $g'(p)\ge 0$ for all $p\in[0,1]$ and therefore the only solution to $g(p)=0$ is $p=0$. The second order of derivative of $g(p)$ is

$$g''(p) = \frac{1}{8\sqrt{2\pi\alpha}\,p^{5/2}}\left(3 - \frac{1}{4\alpha p}\right)\exp\left(-\frac{1}{8\alpha p}\right). \tag{59}$$

If $\alpha > 1/12$, then $g''(p) < 0$ for $p < 1/(12\alpha)$, and $g''(p) > 0$ for $p > 1/(12\alpha) \in (0,1)$. This implies that if $\alpha > 1/12$, there exists an extreme point $p_1 = p(\alpha) > 1/(12\alpha)\in(0,1)$ such that $g'(p_1)=0$ and $g''(p_1)>0$. Meanwhile, as α increases, $g(p)$ monotonically decreases for any $p\in(0,1)$. As $\alpha\to\infty$, $g''(p)>0$ for all $p>0$ and $g(1/2)=0$ but $g(1)>1/2$ for any $\alpha>0$; therefore, $g(p_1)<0$ and $p_1<1/2$. These imply that there must exist $\alpha_0 > 1/12$ such that for $\alpha\ge\alpha_0$, p_1 is the minimum point and $g(p_1)\le 0$. Hence, for $\alpha\ge\alpha_0$, in addition to $p=0$ there exists at least another $p\ge p_0$ such that $g(p)=0$.

In what follows, we outline the numerical solution of α_0 and p_0. According to the above analysis, α_0 and p_0 satisfy both (i) $g'(p_0)=0$ and (ii) $g(p_0)=0$. The former leads to

$$p_0 = \frac{1}{4\sqrt{2\pi\alpha_0 p_0}}\exp\left(-\frac{1}{8\alpha_0 p_0}\right) = \frac{1}{4\sqrt{\pi I/2}}\exp\left(-\frac{1}{2I}\right) \tag{60}$$

where $I = 4\alpha_0 p_0$ is the interference energy. By (ii) we have $p_0 = Q(1/\sqrt{I})$ and then obtain

$$\alpha_0 = \frac{I}{4Q(1/\sqrt{I})}, \tag{61}$$

$$I = \frac{\exp(-1/I)}{8\pi\,Q^2(1/\sqrt{I})}. \tag{62}$$

The interference energy can be solved numerically by iteration

$$I_{n+1} = \frac{\exp(-1/I_n)}{8\pi\,Q^2(1/\sqrt{I_n})}. \tag{63}$$

Afterwards, p_0 and α_0 can be numerically calculated. (Q.E.D.)

Proof of Proposition 2

We first consider an M-class system where the total interference energy is $I = 4\alpha \sum_{j=1}^{M} \lambda_j A_j^2 p_j$ and the BER of the ith class is $p_i = Q(A_i/\sqrt{I})$. Let $g_i(p_i) = p_i - Q(A_i/\sqrt{I}), \forall i$. Similarly to the case of equal-energy distribution, to obtain α^* we shall impose (i) $\partial g_i/\partial p_i = 0$ and (ii) $g_i(p_i) = 0, \forall i$. We have

$$\frac{\partial I}{\partial p_i} = 4\alpha\lambda_i A_i^2 + 4\alpha \sum_{j=1, j\neq i}^{M} \lambda_j A_j^2 \frac{\partial p_j}{\partial p_i} \tag{64}$$

and for $i \neq j$

$$\frac{\partial p_j}{\partial p_i} = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{A_j^2}{2I}\right) A_j \frac{1}{2} I^{-3/2} \frac{\partial I}{\partial p_i}. \tag{65}$$

Hence,

$$\frac{\partial I}{\partial p_i} = 4\alpha\lambda_i A_i^2 + 4\alpha \sum_{j=1, j\neq i}^{M} \lambda_j A_j^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{A_j^2}{2I}\right) \times A_j \frac{1}{2} I^{-3/2} \frac{\partial I}{\partial p_i}. \tag{66}$$

By (i)

$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{A_i^2}{2I}\right) A_i \frac{1}{2} I^{-3/2} \frac{\partial I}{\partial p_i} = 1 \tag{67}$$

which cancels $\partial I/\partial p_i$ jointly with (66) and yields

$$\begin{aligned} &4\alpha\lambda_i A_i^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{A_i^2}{2I}\right) A_i \frac{1}{2} I^{-3/2} \\ &\quad + 4\alpha \sum_{j=1, j\neq i}^{M} \lambda_j A_j^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{A_j^2}{2I}\right) A_j \frac{1}{2} I^{-3/2} \\ &= I^{-3/2} \frac{2\alpha}{\sqrt{2\pi}} \sum_{j=1}^{M} \lambda_j A_j^2 \exp\left(-\frac{A_j^2}{2I}\right) A_j = 1. \end{aligned}$$

Noticing $I^{-1}\alpha = (4\sum_{j=1}^{M} \lambda_j A_j^2 p_j)^{-1}$ and cancelling p_i by (ii), we obtain

$$I = \left[\frac{\sum_{j=1}^{M} \lambda_j A_j^3 \exp\left(-A_j^2/(2I)\right)}{2\sqrt{2\pi} \sum_{j=1}^{M} \lambda_j A_j^2 Q\left(A_j/\sqrt{I}\right)} \right]^2, \tag{68}$$

$$\alpha^* = \frac{I}{4\sum_{i=1}^{M} \lambda_i A_i^2 Q\left(A_i/\sqrt{I}\right)}. \tag{69}$$

Letting $M \to \infty$, we obtain (53) and (54). (Q.E.D.)

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